

ETSI GS QKD 004 push mode specification, KMS server v1.0.0

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Server description of the KMS endpoint processing ETSI GS QKD 004 push mode requests

Base URLs:

- https://qkd_server/api/v1/qkd/etsi004

Web: Support

etsi004

endpoint to push key material to.

POST /key-material

According to the sequence, after a successful `get_key`, the QKD can push the corresponding key material to this endpoint, whenever it created a new key.

Body parameter

```
{
  "status": 0,
  "index": 0,
  "key_buffer": "string",
  "metadata": {
    "size": 57,
    "buffer": {
      "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
    }
  }
}
```

Parameters

Name	In	Type	Required	Description
body	body	object	true	none
» status	body	#/paths/ <code>key-material/post/requestBody/content/application/json/schema/properties</code>	true	none
» »	body	integer	false	Successful
<i>anonymous</i>				
» »	body	integer	false	Successful connection, but peer not connected
<i>anonymous</i>				
» »	body	integer	false	GET_KEY failed because insufficient key available
<i>anonymous</i>				
» »	body	integer	false	GET_KEY failed because peer application is not yet connected
<i>anonymous</i>				
» »	body	integer	false	No QKD connection available
<i>anonymous</i>				
» »	body	integer	false	OPEN_CONNECT failed because the KSID is already in use
<i>anonymous</i>				
» »	body	integer	false	TIMEOUT_ERROR The call failed because the specified TIMEOUT

Name	In	Type	Required	Description
»» <i>anonymous</i>	body	integer	false	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred
»» <i>anonymous</i>	body	integer	false	GET_KEY failed because metadata field size insufficient. Returned Meta-data_size value holds minimum needed size of metadata
» index	body	integer(uint32)	true	increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.
» key_buffer	body	string(base64)	true	none
» metadata	body	object	false	none
»» size	body	integer(uint32)	false	size of buffer in characters
»» buffer	body	object(json)	false	metadata in json format.

Name	In	Type	Required	Description
»»» key_stream_id	body	string(uuid)	true	key stream ID to which the value belongs (required for KMS in case of multiple open key streams)

Example responses

201 Response

```
{
  "status": 0,
  "metadata": {
    "size": 0,
    "buffer": ""
  }
}
```

Responses

Status	Meaning	Description	Schema
201	Created	Key successfully received	Inline

Response Schema

Status Code **201**

Name	Type	Required	Restrictions	Description
» status	#/paths/ 1key-material/post/requestBody/content/application	true	none	noneschema/properties/status(uint32)

oneOf

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	Successful

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	Successful connection, but peer not connected

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	GET_KEY failed because insufficient key available

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	GET_KEY failed because peer application is not yet connected

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	No QKD connection available

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	OPEN_CONNECT failed because the KSID is already in use

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	TIMEOUT_ERROR The call failed because the specified TIMEOUT

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred

xor

Name	Type	Required	Restrictions	Description
» » <i>anonymous</i>	integer	false	none	GET_KEY failed because metadata field size insufficient. Returned Meta- data_size value holds minimum needed size of metadata

continued

Name	Type	Required	Restrictions	Description
» metadata	object	false	none	none
» » size	integer(uint32)	false	none	size of buffer in characters
» » buffer	string(json)	false	none	metadata in json format.

This operation does not require authentication